

THUMRAIT, Oman

WMO: 413140

Lat: 17.666N

Lon: 54.025E

Elev: 1570

StdP: 13.88

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.4	53.4	12.2	10.9	69.4	17.3	13.9	72.1	25.7	74.3	22.9	73.5	6.4	170	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	23.4	108.0	66.1	106.0	66.3	103.9	66.5	76.4	91.0	75.5	90.3	74.8	89.4	11.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.9	129.6	82.3	71.8	124.7	81.7	71.1	121.6	81.4	41.1	90.9	40.3	89.9	39.6	89.9	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.1	29.6	27.1	DB	45.6	111.6	2.4	1.7	43.9	112.8	42.5	113.8	41.2	114.7	39.5	115.9	(4)
(5)			WB	34.7	78.5	2.6	2.8	32.9	80.5	31.3	82.2	29.9	83.8	28.0	85.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.4	66.7	70.9	77.3	84.1	89.1	91.1	87.7	86.5	86.8	81.7	74.6	68.4	(6)	
	DBStd	9.09	4.62	4.98	4.22	3.38	3.13	3.67	4.41	3.99	4.07	3.25	3.09	4.10	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	65	37	11	0	0	0	0	0	0	0	0	0	17	(9)	
	CDD50	11111	516	585	847	1022	1211	1234	1169	1131	1105	983	737	571	(10)	
	CDD65	5701	88	176	382	572	746	784	704	666	655	518	287	123	(11)	
	CDH74	69232	701	1604	3927	7101	10418	11106	8619	7707	8223	6104	2760	962	(12)	
(13)	CDH80	38416	113	516	1831	4062	6410	7008	4798	4219	4961	3365	975	158	(13)	
(14)	Wind	WSAvg	12.9	8.2	10.2	12.5	12.8	13.8	16.6	21.6	20.6	13.4	9.5	7.6	7.4	(14)
(15) Precipitation	PrecAvg	0.9	0.0	0.1	0.1	0.2	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	(15)	
	PrecMax	5.7	0.2	2.3	1.6	4.6	2.9	1.0	0.0	1.5	0.3	0.9	0.4	0.3	(16)	
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	1.5	0.0	0.4	0.4	0.8	0.5	0.2	0.0	0.3	0.0	0.2	0.1	0.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.4	92.1	97.9	102.5	108.2	110.9	109.7	108.4	107.5	100.5	91.6	86.5	(19)	
		MCWB	60.6	61.3	62.7	64.0	65.6	66.5	65.8	68.2	66.3	63.5	61.4	62.1	(20)	
	2%	DB	82.8	88.6	94.7	100.3	105.8	108.5	107.1	104.9	104.8	98.1	89.5	83.7	(21)	
		MCWB	60.9	60.8	61.8	64.3	65.0	65.8	66.1	69.0	66.3	62.9	61.9	62.2	(22)	
	5%	DB	80.6	85.9	91.8	98.3	103.6	106.2	103.1	100.9	102.5	95.9	87.6	81.6	(23)	
		MCWB	61.1	61.0	61.4	63.9	65.2	66.1	68.2	70.9	66.0	62.6	62.1	61.9	(24)	
	10%	DB	78.2	83.0	89.3	96.2	101.2	103.6	98.2	96.8	99.7	93.6	85.7	79.5	(25)	
		MCWB	60.0	60.6	61.6	63.9	65.6	66.8	72.3	72.3	66.3	62.5	62.2	61.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.8	69.3	70.7	74.2	75.8	77.8	77.4	76.2	75.1	73.6	71.1	68.9	(27)	
		MCDB	78.8	81.3	84.2	86.2	89.6	91.9	92.8	93.7	90.6	83.4	79.1	77.6	(28)	
	2%	WB	65.2	66.8	69.2	72.3	74.6	76.5	76.2	75.2	73.6	71.7	69.7	66.5	(29)	
		MCDB	76.5	78.7	81.8	83.8	87.9	90.1	91.5	91.9	89.1	82.1	78.4	75.8	(30)	
	5%	WB	63.8	65.0	68.0	71.1	73.7	75.7	75.3	74.5	72.6	70.2	68.1	65.1	(31)	
		MCDB	74.7	77.1	80.1	83.3	86.4	89.5	90.6	90.7	87.1	80.9	77.4	75.5	(32)	
	10%	WB	62.5	63.5	66.8	70.0	72.9	75.0	74.6	73.8	71.7	68.7	66.4	63.7	(33)	
		MCDB	73.3	76.3	79.4	83.2	85.5	88.8	88.9	89.4	85.1	80.3	77.5	74.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	22.3	23.8	23.4	23.7	23.6	23.4	19.9	20.8	24.3	24.0	22.3	21.7	(35)	
		MCDBR	23.9	26.4	26.4	25.9	27.0	27.3	26.8	26.9	27.2	25.7	23.9	23.2	(36)	
		MCWBR	8.6	9.7	10.1	10.1	10.1	9.8	7.9	6.2	9.0	9.7	9.1	9.0	(37)	
	5% WB	MCDBR	21.1	21.6	19.4	21.6	22.9	22.3	19.5	20.1	23.1	21.2	20.3	20.9	(38)	
		MCWBR	8.0	8.3	6.7	7.0	7.1	6.1	4.6	5.0	6.6	7.9	8.2	8.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.382	0.404	0.452	0.486	0.560	0.707	0.754	0.684	0.546	0.414	0.397	0.384	(40)	
	taud		2.147	2.071	1.933	1.876	1.715	1.461	1.427	1.547	1.771	2.087	2.133	2.144	(41)	
	Ebn at Noon		284	284	273	264	242	207	197	213	244	277	277	279	(42)	
	Edh at Noon		46	52	62	66	76	97	101	90	71	50	46	45	(43)	
(44)	RadAvg		1704	1952	2169	2303	2338	2183	1964	2017	2114	2018	1748	1620	(44)	
(45)	RadStd		45	41	72	60	57	86	86	48	42	48	52	48	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-0.41	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	+0.27	N/A	-2.74	N/A	N/A	-0.54	N/A	N/A	N/A	N/A	

Nomenclature: See separate page